

State-, parameter-, and hybrid-conditioned nonlinear closure maps for projection-based model order reduction

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Abstract

This note revisits explicit higher-mode injection and compares three nonlinear closure strategies for truncated POD coefficients: (i) state-conditioned, $\bar{q} = \mathcal{N}(q)$, (ii) parameter-time conditioned, $\bar{q} = \mathcal{M}(\mu, t)$, and (iii) hybrid, $\bar{q} = \mathcal{H}(q, \mu, t)$. The closure maps are trained from reduced trajectories generated by an n_{tot} -dimensional PROM solve (not from direct projection of HDM snapshots), so the training targets are consistent with the reduced dynamics used online.

All experiments are carried out on the 2D parametric inviscid Burgers benchmark on the 250×250 mesh. We use a two-stage workflow: a baseline training set built from 9 structured parameter points, and an enrichment stage adding 20 Latin-hypercube points (generated with PROM solves). The trained maps are evaluated online with PROM only, and compared against a purely data-driven baseline $q_N(t; \mu) \approx \mathcal{G}(\mu, t)$ and a linear PROM reference in the full truncated space.

Results show that enrichment improves the ANN closures and the purely data-driven surrogate across the test set, while the linear n_{tot} -PROM remains the most accurate reference at a significantly higher online cost. For Case 2, a Petrov-Galerkin (PG) variant improves accuracy but is markedly slower, and the $n = 20$ split highlights a data-limited baseline regime that is strongly corrected after enrichment. A physics-regularized Case 2 PRNN training variant is feasible but remains expensive offline and yields mixed baseline gains. Overall, the PROM-consistent training/evaluation protocol clarifies the comparison between linear and nonlinear reduced formulations.

1 Problem setting and reduced representation

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the high-dimensional state, depending on parameters $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$. Given a snapshot matrix $\mathbf{S} = [\mathbf{u}^1 \cdots \mathbf{u}^{N_s}]$, we define a reference state (global mean)

$$\mathbf{u}_{\text{ref}} := \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{u}^i, \quad (1)$$

and form the centered snapshot matrix $\mathbf{S}_c := [\mathbf{u}^1 - \mathbf{u}_{\text{ref}} \cdots \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}]$. A POD basis $\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N \times n_{\text{tot}}}$ is then constructed from $\mathbf{S}_c$.

We partition the POD basis as

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}], \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \ \bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{n}}, \quad n + \bar{n} = n_{\text{tot}}, \quad (2)$$

where $\mathbf{V}$ spans the retained (primary) subspace and $\bar{\mathbf{V}}$ spans the discarded (secondary) subspace.

With this construction, the standard affine representation for centered states reads

$$\mathbf{u}(t; \boldsymbol{\mu}) - \mathbf{u}_{\text{ref}} \approx \mathbf{V} \mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}} \bar{\mathbf{q}}(t; \boldsymbol{\mu}). \quad (3)$$

2 ROM-consistent training data (critical clarification)

A crucial distinction of the present work is the following:

The reduced coefficients used for training are not obtained by orthogonal projection of high-dimensional trajectories.

Instead, the data is generated through a reduced-order model (ROM) constructed in the full truncated space of dimension n_{tot} . In the present numerical campaign, this ROM is a PROM.

More precisely, let

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \in \mathbb{R}^{n_{\text{tot}}}$$

denote the reduced coordinates obtained by solving the selected n_{tot} -dimensional ROM backend associated with the basis $\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}]$. The n_{tot} -dimensional ROM follows the same time-discrete LSPG formulation used by that backend, so the generated training trajectories are consistent with the corresponding reduced optimality conditions. We decompose these coordinates as

$$\mathbf{q}_N(t; \boldsymbol{\mu}) = \begin{bmatrix} \mathbf{q}(t; \boldsymbol{\mu}) \\ \bar{\mathbf{q}}(t; \boldsymbol{\mu}) \end{bmatrix}, \quad \mathbf{q} \in \mathbb{R}^n, \quad \bar{\mathbf{q}} \in \mathbb{R}^{\bar{n}}, \quad n + \bar{n} = n_{\text{tot}}. \quad (4)$$

For each training parameter value $\boldsymbol{\mu}$, the selected ROM backend is solved in time, yielding trajectories

$$\mathbf{q}(t; \boldsymbol{\mu}), \quad \bar{\mathbf{q}}(t; \boldsymbol{\mu}).$$

We assume that n_{tot} is sufficiently large for this ROM to provide a faithful reduced representation of the HDM dynamics over the parameter domain considered.

The training data for the nonlinear closure maps is extracted from these ROM-consistent coordinates.

Notation simplification. From this point onward, the subscript N will be omitted, and all reduced coordinates are understood to be those obtained from the n_{tot} -dimensional ROM unless explicitly stated otherwise.

Thus, the learned mappings approximate relations between ROM-generated reduced coordinates,

$$\mathbf{q}(t; \boldsymbol{\mu}) \quad \text{and} \quad \bar{\mathbf{q}}(t; \boldsymbol{\mu}),$$

rather than the projection coefficients of the high-dimensional solution.

3 Nonlinear manifold formulation and closure variants

Having established that the training data is generated from ROM-consistent reduced coordinates $(\mathbf{q}, \bar{\mathbf{q}})$, we now introduce a nonlinear trial manifold in which the secondary coordinates are represented through a closure map.

In the n_{tot} -dimensional ROM used for data generation, both $\mathbf{q}$ and $\bar{\mathbf{q}}$ are solved online. Here, we instead construct an n -dimensional nonlinear manifold ROM in which only $\mathbf{q}$ remains as an online unknown, while the truncated coordinates are reconstructed through a learned map. Specifically, we approximate

$$\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{F}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}, t), \quad (5)$$

where the functional dependence of $\mathcal{F}$ determines the structure of the nonlinear manifold.

The corresponding reduced approximation of the state is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}} \mathcal{F}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}, t). \quad (6)$$

In an LSPG or Gauss–Newton framework, the Jacobian of the manifold with respect to the retained coordinates reads

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}}(\mathbf{q}, \boldsymbol{\mu}, t). \quad (7)$$

Thus, the influence of the closure on the reduced optimality system is determined by the dependence of $\mathcal{F}$ on $\mathbf{q}$. In particular, state-dependent closures contribute the term $\bar{\mathbf{V}} \partial \mathcal{F} / \partial \mathbf{q}$ to the manifold tangent, whereas purely parameter–time closures yield the constant tangent $\mathbf{T} = \mathbf{V}$.

Time-discrete LSPG problem solved online Let the high-dimensional residual be defined as

$$\mathbf{r}(\mathbf{u}, t; \boldsymbol{\mu}) = \mathbf{M}(\boldsymbol{\mu}) \dot{\mathbf{u}} + \mathbf{f}(\mathbf{u}; \boldsymbol{\mu}) - \mathbf{g}(t; \boldsymbol{\mu}). \quad (8)$$

Let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the time-discrete residual of the HDM at step $m+1$. Given a nonlinear trial manifold

$$\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1}; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}^{m+1} + \bar{\mathbf{V}} \mathcal{F}(\mathbf{q}^{m+1}, \boldsymbol{\mu}, t^{m+1}),$$

the online reduced unknown at each time step is the retained coordinate vector $\mathbf{q}^{m+1} \in \mathbb{R}^n$.

In all three closure-based cases, the reduced state is computed by a least-squares Petrov–Galerkin (LSPG) solve:

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}); \boldsymbol{\mu})\|_2^2. \quad (9)$$

This nonlinear least-squares problem is solved by Gauss–Newton iterations. Let $\mathbf{J}^{m+1,k} := \partial \mathbf{r}^{m+1} / \partial \mathbf{u}$ be the residual Jacobian evaluated at the current iterate, and define the manifold tangent

$$\mathbf{T}^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}) = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}}(\mathbf{q}^{m+1,k}, \boldsymbol{\mu}, t^{m+1}).$$

Then the LSPG test basis is

$$\mathbf{W}^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}^{m+1,k},$$

and the Gauss–Newton update $\Delta \mathbf{q}^{m+1,k}$ solves

$$(\mathbf{W}^{m+1,k})^T \mathbf{W}^{m+1,k} \Delta \mathbf{q}^{m+1,k} = -(\mathbf{W}^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (10)$$

The three closure variants differ only through the choice of $\mathcal{F}$ and thus through the manifold tangent $\mathbf{T}^{m+1,k}$.

The purely data-driven baseline discussed later does not solve this online optimization problem; it predicts reduced coordinates directly from $(\boldsymbol{\mu}, t)$.

Hyperreduced formulation used in practice Although the formulation above defines the nonlinear manifold ROM, its direct implementation still requires evaluating the residual $\mathbf{r}^{m+1}$ and Jacobian $\mathbf{J}^{m+1,k}$ at the full dimension N . Consequently, the online cost remains tied to the HDM.

When the HPROM backend is used, we solve a hyperreduced version of this problem based on the energy-conserving sampling and weighting (ECSW) method. The key idea is to approximate the residual by restricting its evaluation to a reduced subset of mesh entities.

Let $\mathcal{E} = \{e_i\}_{i=1}^{N_e}$ denote the set of mesh entities. We assume an element-wise decomposition

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ \mathbf{u}; \boldsymbol{\mu}). \quad (11)$$

ECSW replaces this sum by a weighted sum over a reduced subset $\tilde{\mathcal{E}} \subset \mathcal{E}$:

$$\tilde{\mathbf{r}}^{m+1}(\mathbf{q}; \boldsymbol{\mu}) = \sum_{e_i \in \tilde{\mathcal{E}}} \xi_{e_i} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i} \tilde{\mathbf{u}}^{m+1}(\mathbf{q}; \boldsymbol{\mu}); \boldsymbol{\mu}), \quad (12)$$

with $\xi_{e_i} > 0$.

The online problem actually solved is therefore

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (13)$$

Defining the hyperreduced Jacobian

$$\tilde{\mathbf{J}}^{m+1,k} := \left. \frac{\partial \tilde{\mathbf{r}}^{m+1}}{\partial \mathbf{u}} \right|_{\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu})},$$

the Gauss–Newton iterations proceed as

$$\begin{aligned} \widetilde{\mathbf{W}}^{m+1,k} &= \tilde{\mathbf{J}}^{m+1,k} \mathbf{T}^{m+1,k}, \\ (\widetilde{\mathbf{W}}^{m+1,k})^T \widetilde{\mathbf{W}}^{m+1,k} \Delta \mathbf{q}^{m+1,k} &= -(\widetilde{\mathbf{W}}^{m+1,k})^T \tilde{\mathbf{r}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}). \end{aligned} \quad (14)$$

Importantly, hyperreduction modifies only the residual and Jacobian evaluations, while the nonlinear manifold and its tangent remain unchanged. Thus, all closure variants are treated within the same ECSW framework.

We now specialize this general formulation into three distinct variants.

Case 1: State-dependent closure.

In the first variant, the secondary coordinates depend only on the retained ones,

$$\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{N}(\mathbf{q}(t; \boldsymbol{\mu})). \quad (15)$$

The manifold becomes

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q}), \quad (16)$$

and the Jacobian reads

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}}{\partial \mathbf{q}}.$$

This formulation represents the nonlinear manifold as a graph over the retained subspace. It implicitly assumes that the mapping $\mathbf{q} \mapsto \bar{\mathbf{q}}$ is (approximately) single-valued over the region explored by the dynamics. When this injectivity condition deteriorates, the learned closure represents an effective approximation of potentially branch-dependent secondary content.

Case 2: Parameter–time dependent closure.

In the second variant, the secondary coordinates are modeled directly as functions of parameters and time,

$$\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{M}(\boldsymbol{\mu}, t). \quad (17)$$

The reduced manifold becomes

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{M}(\boldsymbol{\mu}, t), \quad (18)$$

and the Jacobian simplifies to

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

In this case, the higher-mode contribution is injected explicitly and does not depend on the evolving reduced state. Consequently, within Gauss–Newton iterations, the secondary component is prescribed by $(\boldsymbol{\mu}, t)$ and does not adjust in response to changes in $\mathbf{q}$.

Case 2 with enriched residual testing (Petrov–Galerkin, PG).

The standard Case 2 formulation retains the parameter–time closure

$$\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{M}(\boldsymbol{\mu}, t),$$

so that the online unknown remains only the retained coordinate vector $\mathbf{q}^{m+1} \in \mathbb{R}^n$, and the trial manifold at step $m+1$ is

$$\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1}; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}^{m+1} + \bar{\mathbf{V}} \mathcal{M}(\boldsymbol{\mu}, t^{m+1}). \quad (19)$$

Since the prescribed secondary component does not depend on $\mathbf{q}^{m+1}$, the manifold tangent is

$$\mathbf{T}^{m+1,k} = \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}} = \mathbf{V}.$$

A structural limitation of standard LSPG in this setting is that the residual is tested only through $\mathbf{J}^{m+1,k} \mathbf{V}$, although the state itself contains the injected higher-mode contribution $\bar{\mathbf{V}} \mathcal{M}(\boldsymbol{\mu}, t^{m+1})$. To strengthen the residual criterion while still solving only for $\mathbf{q}^{m+1}$, we introduce the full truncated basis

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}] \in \mathbb{R}^{N \times n_{\text{tot}}}, \quad n_{\text{tot}} = n + \bar{n},$$

and define the projected residual

$$\mathbf{s}^{m+1}(\mathbf{q}; \boldsymbol{\mu}) := (\mathbf{J}^{m+1}(\mathbf{q}; \boldsymbol{\mu}) \mathbf{V}_{\text{tot}})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}; \boldsymbol{\mu}); \boldsymbol{\mu}) \in \mathbb{R}^{n_{\text{tot}}}. \quad (20)$$

The online reduced state is then defined by the least-squares problem

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \frac{1}{2} \|\mathbf{s}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (21)$$

This formulation should not be interpreted as a square Petrov–Galerkin system with left basis $\mathbf{J}^{m+1} \mathbf{V}_{\text{tot}}$, since $\mathbf{s}^{m+1} \in \mathbb{R}^{n_{\text{tot}}}$ whereas the online unknown remains $\mathbf{q}^{m+1} \in \mathbb{R}^n$. Rather, it is an overdetermined projected-residual minimization in which the trial updates remain confined to $\text{Range}(\mathbf{V})$, while the residual is tested through the richer space associated with $\mathbf{V}_{\text{tot}}$.

Gauss–Newton iterations. Let

$$\mathbf{J}^{m+1,k} := \left. \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}} \right|_{\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu})}$$

be the residual Jacobian at the current iterate, and define

$$\mathbf{s}^{m+1,k} := (\mathbf{J}^{m+1,k} \mathbf{V}_{\text{tot}})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}).$$

Neglecting derivatives of $\mathbf{J}^{m+1}$ with respect to $\mathbf{q}$ in the Gauss–Newton approximation yields

$$\left. \frac{\partial \mathbf{s}^{m+1}}{\partial \mathbf{q}} \right|_{\mathbf{q}^{m+1,k}} \approx (\mathbf{J}^{m+1,k} \mathbf{V}_{\text{tot}})^T \mathbf{J}^{m+1,k} \mathbf{V}. \quad (22)$$

Introducing

$$\mathbf{Z}^{m+1,k} := (\mathbf{J}^{m+1,k} \mathbf{V}_{\text{tot}})^T \mathbf{J}^{m+1,k} \mathbf{V} \in \mathbb{R}^{n_{\text{tot}} \times n}, \quad (23)$$

the Gauss–Newton update $\Delta \mathbf{q}^{m+1,k} \in \mathbb{R}^n$ solves

$$(\mathbf{Z}^{m+1,k})^T \mathbf{Z}^{m+1,k} \Delta \mathbf{q}^{m+1,k} = -(\mathbf{Z}^{m+1,k})^T \mathbf{s}^{m+1,k}. \quad (24)$$

This variant leaves the Case 2 trial manifold unchanged and therefore does not make the secondary coordinates into online unknowns. In particular, the trial tangent remains

$$\frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}} = \mathbf{V},$$

so the Gauss–Newton updates are still restricted to directions in $\text{Range}(\mathbf{V})$. Consequently, this enriched-testing variant does not remove the structural limitation of standard Case 2, namely that the prescribed higher-mode contribution $\bar{\mathbf{V}}\mathcal{M}(\boldsymbol{\mu}, t^{m+1})$ enters the residual value but not the tangent directions explored by the online minimization. Its role is more limited: it strengthens the residual criterion by penalizing residual components seen through the full truncated space $\mathbf{V}_{\text{tot}}$, while preserving an n -dimensional online solve.

Case 3: Hybrid state–parameter closure.

The third variant combines both sources of information,

$$\bar{\mathbf{q}}(t; \boldsymbol{\mu}) = \mathcal{H}(\mathbf{q}(t; \boldsymbol{\mu}), \boldsymbol{\mu}, t). \quad (25)$$

The manifold reads

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t), \quad (26)$$

with Jacobian

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{H}}{\partial \mathbf{q}}.$$

This hybrid formulation restores state-dependent coupling while allowing the closure to incorporate parameter and temporal information. It is strictly more expressive than Cases 1 and 2 in terms of hypothesis space, at the cost of a higher-dimensional regression problem and potentially more delicate conditioning.

4 Model problem: 2D parametric inviscid Burgers benchmark

We consider the 2D parametric inviscid Burgers initial-boundary-value problem (IBVP) introduced in [?], which serves as a nonlinear transport-dominated benchmark with shock formation. The governing equations are

$$\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} = 0.02 \exp(\mu_2 x), \quad (27)$$

$$\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} = 0, \quad (28)$$

with parameter vector $\boldsymbol{\mu} = (\mu_1, \mu_2)$, spatial domain $\Omega = [0, 100] \times [0, 100]$, and final time $T_f = 25$.

The boundary and initial conditions are

$$u_x(0, y, t; \boldsymbol{\mu}) = \mu_1, \quad (29)$$

$$u_x(x, y, 0) = 1, \quad u_y(x, y, 0) = 1. \quad (30)$$

Parameter μ_1 controls the left boundary value of u_x , while μ_2 enters the exponential source term in (27). Variations in μ_2 significantly affect the location and strength of shock structures, making the problem strongly nonlinear over the parameter domain

$$\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]. \quad (31)$$

Spatial and temporal discretization. The system is semi-discretized using a cell-centered finite-volume Godunov scheme. For a structured mesh with $N_x \times N_y$ cells, the state dimension is

$$N = 2 N_x N_y.$$

Time integration employs 500 uniform steps of size $\Delta t = 0.05$ using the midpoint rule. At each step, the resulting nonlinear algebraic system is solved using a direct sparse solver for the high-dimensional model (HDM).

The reduced models are constructed using the same time-discrete residual as the HDM and are solved using an LSPG/Gauss–Newton formulation projected onto the corresponding reduced trial manifold.

Although the workflow is portable to other meshes, the numerical campaign reported in this manuscript is restricted to:

- A 250×250 mesh ($N = 125,000$).

Parameter sampling for snapshot generation

The baseline training set is sampled using a uniform 3×3 tensor grid, producing 9 training parameter instances. Figure 1 shows this baseline design together with the three evaluation points used in the report. For each parameter point, 501 snapshots (including the initial condition) are collected, yielding a total of $N_s = 4509$ snapshots.

These HDM snapshots are first used to construct the POD basis $\mathbf{V}_{\text{tot}}$. Subsequently, for each training parameter value, an n_{tot} -dimensional ROM backend is solved in time and the corresponding reduced trajectories $(\mathbf{q}(t; \boldsymbol{\mu}), \bar{\mathbf{q}}(t; \boldsymbol{\mu}))$ are extracted. These ROM-consistent reduced coefficients constitute the dataset used to train the nonlinear closure maps. In the present numerical campaign (Section 5), the training trajectories are generated with PROM.

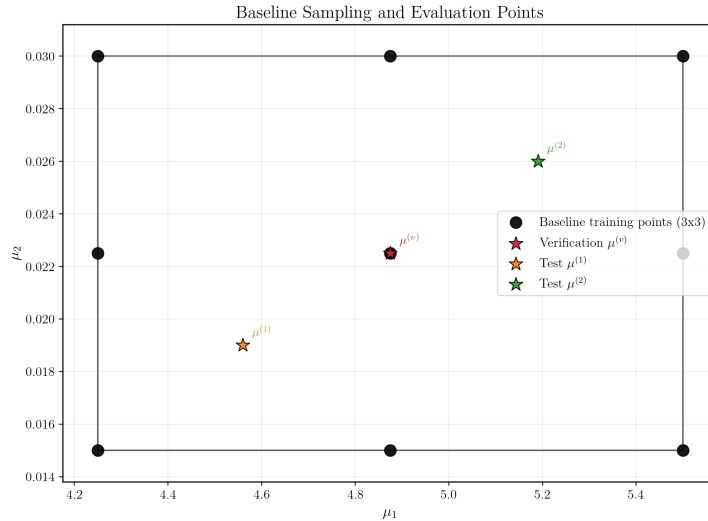


Figure 1: Baseline 3×3 structured sampling points in $\mathcal{D}$ (black circles) and evaluation points $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, $\boldsymbol{\mu}^{(2)}$ (colored stars).

5 PROM workflow and evaluation on the 250×250 mesh

Scope. All results in this section are based on the 250×250 mesh only ($N = 125,000$). The nonlinear maps are trained from PROM-generated reduced trajectories and evaluated online with PROM only. The HPROM/ECSW route remains available in the implementation and

is described in the methodological section, but it is not used in the quantitative comparisons reported below. We do not include oracle diagnostics as table rows in this version, except for a brief implementation sanity remark for the PG variant.

End-to-end pipeline. We follow the sequence below:

1. Build a POD basis from HDM snapshots at 9 equidistant training parameters (uniform 3×3 grid on $\mathcal{D}$).
2. Retain $n_{\text{tot}} = 151$ modes using tolerance-based truncation (`pod_tol` = 10^{-4}), with $n = 10$ primary and $\bar{n} = 141$ secondary modes.
3. Build the linear reduced model in the full space $\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N \times 151}$ and generate the Stage-2 coefficient dataset from PROM trajectories:

$$\mathbf{q}_N^{\text{PROM}}(t; \boldsymbol{\mu}) = \begin{bmatrix} \mathbf{q}(t; \boldsymbol{\mu}) \\ \bar{\mathbf{q}}(t; \boldsymbol{\mu}) \end{bmatrix}.$$

4. Train four nonlinear maps from this dataset: Case 1, Case 2, Case 3, and the purely data-driven map $\mathcal{G}(\boldsymbol{\mu}, t) \approx \mathbf{q}_N^{\text{PROM}}$. In addition, we test a Case 2 PRNN variant with combined data and projected-residual loss.
5. Evaluate all models at three points (verification point first, then two off-grid test points):

$$\boldsymbol{\mu}^{(v)} = (4.875, 0.0225), \quad \boldsymbol{\mu}^{(1)} = (4.56, 0.019), \quad \boldsymbol{\mu}^{(2)} = (5.19, 0.026),$$

using PROM online solves.

6. Enrichment stage: add 20 LHS points, regenerate PROM coefficients $\mathbf{q}_N^{\text{PROM}}$ (no HDM reruns), retrain the same four maps, and re-evaluate at the same three points.

Data-driven baseline (very brief). The data-driven model is a direct regression

$$\mathcal{G} : (\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N(t; \boldsymbol{\mu}),$$

that predicts all n_{tot} coefficients from parameter and time only. It does not run an online LSPG/Gauss–Newton solve.

ANN architecture and optimization setup. For the production runs in this report (Case 1, Case 2, Case 3, and data-driven), we use the same MLP core architecture with ELU activations:

$$32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 256,$$

followed by an output layer of task-dependent size. Training uses AdamW with learning rate 10^{-3} , weight decay 10^{-6} , early stopping (patience 120), and fixed seed 42. Case 1, Case 2, Case 3, and data-driven use batch size 128. For Case 2 with $n = 20$, we also tested several alternative architectures/activations and optimizer settings; the best run in our tests still used this same baseline architecture.

Case 2 PRNN training (offline). For the Case 2 closure map $\bar{\mathbf{q}} = \mathcal{M}_\theta(\boldsymbol{\mu}, t)$, we tested a physics-regularized training objective:

$$\mathcal{L}(\theta) = \omega_{\text{data}} \mathcal{L}_{\text{data}}(\theta) + \omega_{\text{res}} \mathcal{L}_{\text{res}}(\theta).$$

The data term is

$$\mathcal{L}_{\text{data}} = \frac{1}{N_b} \sum_{j=1}^{N_b} \left\| \mathcal{M}_\theta(\boldsymbol{\mu}_j, t_j) - \bar{\mathbf{q}}_j^{\text{ref}} \right\|_2^2.$$

The physics term is built from a projected HDM residual:

$$\hat{\mathbf{u}}_j = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}_j^{\text{ref}} + \bar{\mathbf{V}} \mathcal{M}_\theta(\boldsymbol{\mu}_j, t_j), \quad \mathcal{L}_{\text{res}} = \frac{1}{N_{\text{phys}}} \sum_{j \in \mathcal{I}_{\text{phys}}} \frac{1}{k} \left\| \mathbf{P}^T \mathbf{r}(\hat{\mathbf{u}}_j; \boldsymbol{\mu}_j) \right\|_2^2,$$

with $\mathbf{P} \in \{\mathbf{V}_{\text{tot}}, \mathbf{V}\}$ and $k = \dim(\text{Range}(\mathbf{P}))$. In this campaign we use $\mathbf{P} = \mathbf{V}_{\text{tot}}$. Using the available residual Jacobian $\mathbf{J}_j = \partial \mathbf{r} / \partial \mathbf{u}|_{\hat{\mathbf{u}}_j}$, the residual term contributes

$$\frac{\partial \mathcal{L}_{\text{res}}}{\partial \bar{\mathbf{q}}_j} \approx \frac{2}{k} \bar{\mathbf{V}}^T \mathbf{J}_j^T \mathbf{P} (\mathbf{P}^T \mathbf{r}_j).$$

To keep training affordable, we use a two-stage schedule (data-only pretraining, then short PRNN fine-tuning), stochastic subsampling of physics rows (`physics_subsample`), and optional sparse evaluation frequency (`physics_every`). Even with these controls, PRNN training remains substantially more expensive than pure data training for Case 2.

Dataset sizes. Baseline Stage-3 training uses $M = 4509$ samples (9×501). Enriched Stage-3 training uses $M = 14529$ samples $((9 + 20) \times 501)$.

POD spectrum. Figure 2 reports the singular-value decay used to select the POD truncation at tolerance `pod_tol` = 10^{-4} . Figure 3 shows a zoom on the first 50 modes.

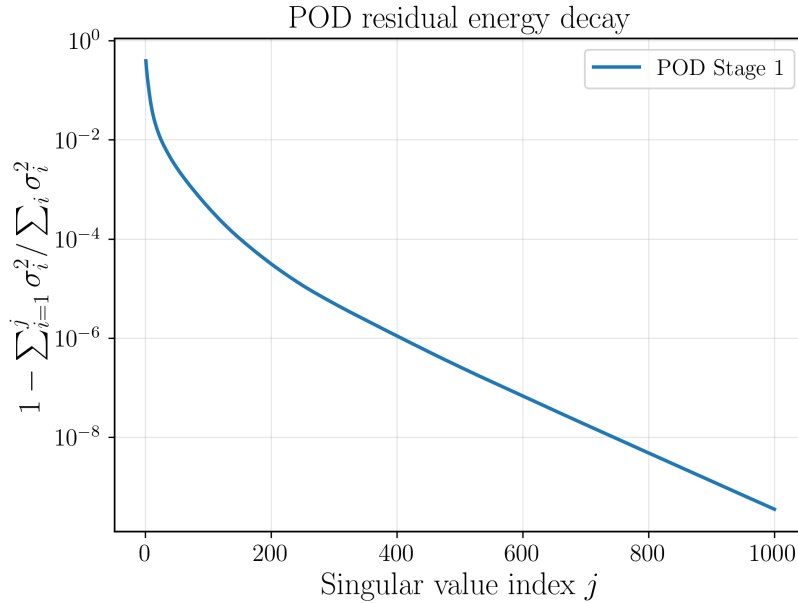


Figure 2: Singular value decay of the centered HDM snapshot matrix on the 250×250 mesh (Stage 1 POD).

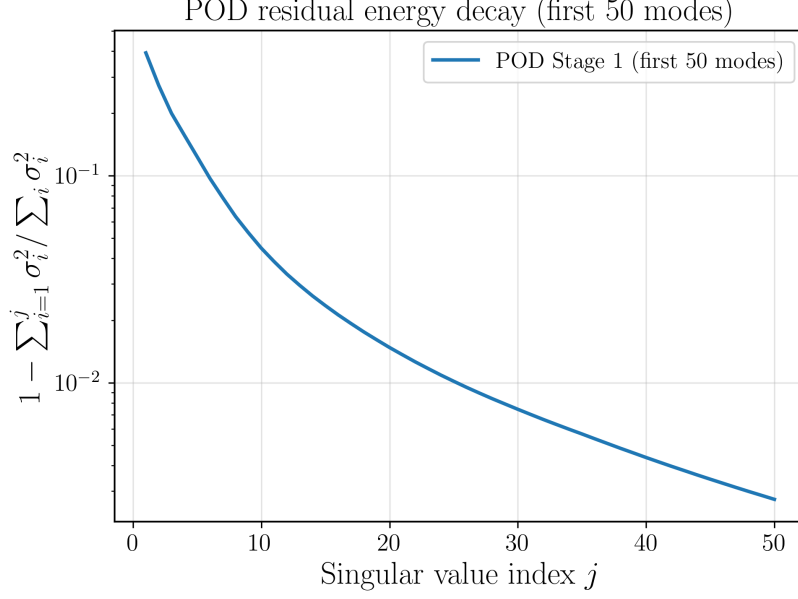


Figure 3: Zoom of the Stage 1 POD singular-value decay for the first 50 modes on the 250×250 mesh.

Sampling used for enrichment. Figure 4 shows the original 3×3 training points together with the 20 LHS enrichment points used to build the enriched PROM coefficient dataset, plus the three evaluation points.

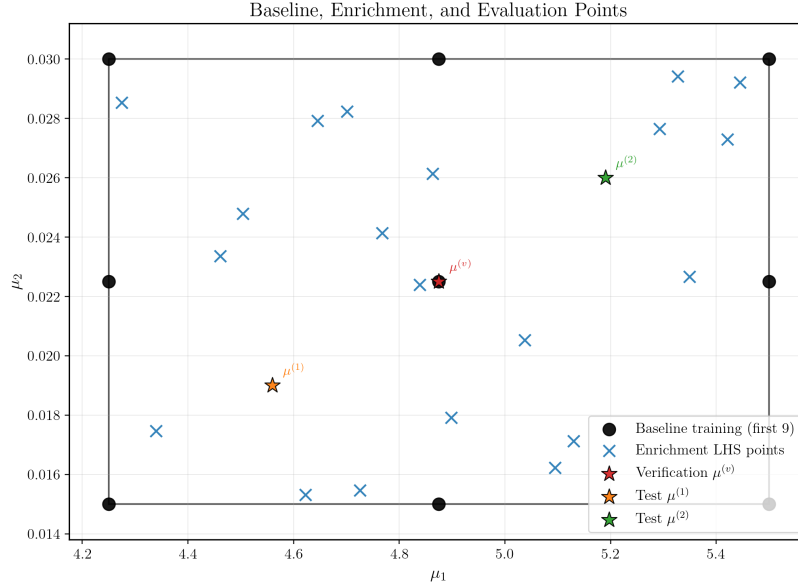


Figure 4: Baseline training points (9 equidistant points), enrichment points (20 LHS samples), and evaluation points $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$ over $\mathcal{D}$.

5.1 Baseline models (9-point training dataset)

Table 1 reports trajectory-level relative errors and online runtimes for the PROM-consistent campaign. We include two linear references (the main $n_{\text{tot}} = 151$ PROM and an additional linear PROM with $n_{\text{tot}} = 76$), the three ANN closure variants, the Case 2 Petrov-Galerkin

(PG) oracle diagnostic, a Case 2 PRNN variant (same Case 2 map with projected-residual regularization in training), two Case 2 $n = 20$ baseline variants (native $n = 20$ training and $n = 10 \rightarrow n = 20$ trimmed transfer), and the purely data-driven model. The HPROM backend remains available in the codebase, but it is not used for the quantitative tables in this section. In all result tables below, “Case 2 + PG Oracle” denotes the *oracle-injected Case 2 PG* run (secondary coordinates injected from linear n_{tot} -PROM). The non-oracle PG variant is not included in this version of the report. The trajectory-level relative error is computed against the HDM trajectory:

$$\text{RelErr}(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu}) - \mathbf{U}_{\text{model}}(\boldsymbol{\mu})\|_F}{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu})\|_F},$$

where $\mathbf{U} \in \mathbb{R}^{N \times (N_t+1)}$ stacks the state snapshots in time.

Table 1: Baseline PROM results on 250×250 ($n_{\text{tot}} = 151$), including Case 2 variants for $n = 10$ and $n = 20$.

Method	RelErr($\boldsymbol{\mu}^{(v)}$) [%]	RelErr($\boldsymbol{\mu}^{(1)}$) [%]	RelErr($\boldsymbol{\mu}^{(2)}$) [%]	Time($\boldsymbol{\mu}^{(v)}$) [s]	Time($\boldsymbol{\mu}^{(1)}$) [s]	Time($\boldsymbol{\mu}^{(2)}$) [s]
Linear PROM ($n_{\text{tot}} = 151$)	0.410	0.450	0.463	921.943	871.356	913.506
Linear PROM ($n_{\text{tot}} = 76$)	1.314	1.574	1.398	449.279	450.632	451.384
Case 1 ANN ($\mathbf{q} = \mathcal{N}(\mathbf{q})$)	0.708	0.716	1.103	148.840	147.333	149.762
Case 2 ANN ($\mathbf{q} = \mathcal{M}(\boldsymbol{\mu}, t)$)	0.874	1.224	1.507	90.985	92.647	91.022
Case 2 ANN + PG Oracle (enriched residual testing)	0.793	1.202	1.223	409.562	409.104	406.139
Case 2 PRNN (two-stage, projected residual regularization)	0.999	1.478	1.150	158.195	163.423	159.270
Case 2 ANN ($n = 20$, $\bar{n} = 131$, native $n = 20$ model)	0.652	6.502	3.755	146.334	141.090	141.372
Case 2 ANN ($n = 20$, $\bar{n} = 131$, $n = 10 \rightarrow n = 20$ trimmed transfer)	0.689	1.056	1.035	140.073	143.743	143.484
Case 3 ANN ($\mathbf{q} = \mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t)$)	0.655	0.765	0.788	149.478	149.438	151.618
Data-driven ($\mathbf{q}_N = \mathcal{G}(\boldsymbol{\mu}, t)$)	0.874	1.832	1.580	0.005	0.005	0.005

Baseline $n = 20$ transfer note. The second $n = 20$ Case 2 baseline row is a transfer run that reuses the baseline $n = 10$ Case 2 checkpoint and trims the first 10 secondary outputs (modes 11–20), so the solver uses the $n = 20$ split with $\bar{n} = 131$. Compared with the direct baseline native- $n = 20$ checkpoint, this reduces off-grid errors from 6.502% $\rightarrow$ 1.056% at $\boldsymbol{\mu}^{(1)}$ and from 3.755% $\rightarrow$ 1.035% at $\boldsymbol{\mu}^{(2)}$, while the verification-point error changes from 0.652% $\rightarrow$ 0.689%.

5.2 Enriched models (9+20 training dataset)

For enrichment, we augment the baseline 9 trajectories with 20 LHS trajectories and retrain all four nonlinear maps. Table 2 reports PROM results at the same three points, including the Case 2 PG Oracle and $n = 20$ variants. The linear reference rows are unchanged and are included for reference. The Case 2 PRNN row is marked as not run in the current enriched campaign.

Table 2: Enriched PROM results on 250×250 (trained with 9 + 20 trajectories, $n_{\text{tot}} = 151$), including Case 2 variants for $n = 10$ and $n = 20$.

Method	RelErr($\boldsymbol{\mu}^{(v)}$) [%]	RelErr($\boldsymbol{\mu}^{(1)}$) [%]	RelErr($\boldsymbol{\mu}^{(2)}$) [%]	Time($\boldsymbol{\mu}^{(v)}$) [s]	Time($\boldsymbol{\mu}^{(1)}$) [s]	Time($\boldsymbol{\mu}^{(2)}$) [s]
Linear PROM ($n_{\text{tot}} = 151$)	0.410	0.450	0.463	921.943	871.356	913.506
Linear PROM ($n_{\text{tot}} = 76$)	1.314	1.574	1.398	449.279	450.632	451.384
Case 1 ANN (enriched)	0.681	0.561	0.922	156.034	156.136	156.006
Case 2 ANN (enriched)	0.894	0.962	0.875	91.416	91.013	90.763
Case 2 ANN + PG Oracle (enriched)	0.685	0.754	0.714	413.579	411.149	415.588
Case 2 PRNN (enriched)	–	–	–	–	–	–
Case 2 ANN ($n = 20$, $\bar{n} = 131$, enriched)	0.499	0.550	0.548	140.796	141.836	141.215
Case 3 ANN (enriched)	0.473	0.517	0.532	156.095	155.333	165.269
Data-driven (enriched)	0.860	0.928	0.896	0.006	0.006	0.006

5.3 Runtime and speed-up reporting

Using measured HDM wall-clock times

$$t_{\text{HDM}}(\boldsymbol{\mu}^{(v)}) = 7.542 \times 10^2 \text{ s}, \quad (32)$$

$$t_{\text{HDM}}(\boldsymbol{\mu}^{(1)}) = 7.507 \times 10^2 \text{ s}, \quad (33)$$

$$t_{\text{HDM}}(\boldsymbol{\mu}^{(2)}) = 7.577 \times 10^2 \text{ s}, \quad (34)$$

where $t_{\text{HDM}}(\boldsymbol{\mu}^{(v)})$ is taken as a credible interpolation-level reference between the two measured off-grid HDM times. we define speed-up at each test point as

$$S_m(\boldsymbol{\mu}) = \frac{t_{\text{HDM}}(\boldsymbol{\mu})}{t_m(\boldsymbol{\mu})}.$$

Table 3 reports baseline and enriched speed-ups for all three points.

Table 3: PROM speed-up factors relative to HDM at the three reported points.

Method	Baseline $S(\boldsymbol{\mu}^{(v)})$	Baseline $S(\boldsymbol{\mu}^{(1)})$	Baseline $S(\boldsymbol{\mu}^{(2)})$	Enriched $S(\boldsymbol{\mu}^{(v)})$	Enriched $S(\boldsymbol{\mu}^{(1)})$	Enriched $S(\boldsymbol{\mu}^{(2)})$
Linear PROM ($n_{\text{tot}} = 151$)	0.82	0.86	0.83	0.82	0.86	0.83
Linear PROM ($n_{\text{tot}} = 76$)	1.68	1.67	1.68	1.68	1.67	1.68
Case 1 ANN	5.07	5.10	5.06	4.83	4.81	4.86
Case 2 ANN	8.29	8.10	8.32	8.25	8.25	8.35
Case 2 ANN + PG Oracle	1.84	1.84	1.87	1.82	1.83	1.82
Case 2 PRNN	4.77	4.59	4.76	—	—	—
Case 2 ANN ($n = 20$, native)	5.15	5.32	5.36	5.36	5.29	5.37
Case 2 ANN ($n = 20$, trimmed transfer)	5.38	5.22	5.28	—	—	—
Case 3 ANN	5.05	5.02	5.00	4.83	4.83	4.58
Data-driven	1.44×10^5	1.43×10^5	1.46×10^5	1.20×10^5	1.18×10^5	1.34×10^5

Note. The data-driven runtime is pure ANN inference time and does not include a reduced nonlinear solve. Therefore, its speed-up values are not directly comparable to PROM nonlinear-solve timings.

5.4 Effect of enrichment

Define relative error reduction from baseline to enriched as

$$\Delta e = 100 \frac{e_{\text{base}} - e_{\text{enriched}}}{e_{\text{base}}}.$$

Table 4 reports baseline and enriched errors together with Δe at all three points.

Table 4: Baseline vs enriched trajectory errors and relative reduction (%) on 250×250 . Positive Δe indicates improvement.

Method	$e_{\text{base}}(\boldsymbol{\mu}^{(v)})$ [%]	$e_{\text{enr}}(\boldsymbol{\mu}^{(v)})$ [%]	$\Delta e(\boldsymbol{\mu}^{(v)})$ [%]	$e_{\text{base}}(\boldsymbol{\mu}^{(1)})$ [%]	$e_{\text{enr}}(\boldsymbol{\mu}^{(1)})$ [%]	$\Delta e(\boldsymbol{\mu}^{(1)})$ [%]	$e_{\text{base}}(\boldsymbol{\mu}^{(2)})$ [%]	$e_{\text{enr}}(\boldsymbol{\mu}^{(2)})$ [%]	$\Delta e(\boldsymbol{\mu}^{(2)})$ [%]
Linear PROM ($n_{\text{tot}} = 151$)	0.410	0.410	0.00	0.450	0.450	0.00	0.463	0.463	0.00
Linear PROM ($n_{\text{tot}} = 76$)	1.314	1.314	0.00	1.574	1.574	0.00	1.398	1.398	0.00
Case 1 ANN	0.708	0.681	3.75	0.716	0.561	21.68	1.103	0.922	16.43
Case 2 ANN	0.874	0.894	-2.36	1.224	0.962	21.39	1.507	0.875	41.96
Case 2 ANN + PG Oracle	0.793	0.685	13.56	1.202	0.754	37.28	1.223	0.714	41.59
Case 2 PRNN	0.999	—	—	1.478	—	—	1.150	—	—
Case 2 ANN ($n = 20$)	0.652	0.499	23.56	6.502	0.550	91.54	3.755	0.548	85.40
Case 3 ANN	0.655	0.473	27.75	0.765	0.517	32.45	0.788	0.532	32.44
Data-driven	0.874	0.860	1.67	1.832	0.928	49.35	1.580	0.896	43.28

Remark and practical takeaway. In this PROM-consistent campaign, the linear $n_{\text{tot}} = 151$ PROM remains the most accurate reference but at the highest online cost. The additional linear run with $n_{\text{tot}} = 76$ reduces linear runtime to about 450 s but with larger trajectory errors (about 1.3–1.6%). For Case 2, the PG Oracle row improves all three enriched-point errors relative to the standard Case 2 row, but with a substantially larger runtime (speed-up near $1.8\times$ only). The PRNN-trained Case 2 row shows mixed baseline impact (improvement at $\boldsymbol{\mu}^{(2)}$, degradation

at $\mu^{(v)}$ and $\mu^{(1)}$) and is slower online than standard Case 2 (about $4.6\text{--}4.8\times$ speed-up vs about $8.1\text{--}8.3\times$), so it is not a dominant tradeoff in the present setting. For the $n = 20$ Case 2 split, we report both baseline variants: the native $n = 20$ model and a baseline transfer run ($n = 10 \rightarrow n = 20$ trimming). The transfer run yields much better off-grid baseline behavior than the native baseline model (from $6.502\% \rightarrow 1.056\%$ at $\mu^{(1)}$ and $3.755\% \rightarrow 1.035\%$ at $\mu^{(2)}$), while the enriched native- $n = 20$ model already achieves low errors (about 0.55%) at both off-grid points.

PG oracle sanity check (implementation). As a temporary implementation check, we ran the Case 2 PG solve using an oracle truncated trajectory $\bar{\mathbf{q}}(t)$ taken directly from the linear n_{tot} -PROM (instead of ANN prediction). On the verification and test points, this oracle-PG run converges to the linear PROM solution within printed precision: relative error vs linear PROM is 0.000% , relative primary-coordinate mismatch is $0.000\%\text{--}0.001\%$, and relative truncated-coordinate mismatch is 0.000% . The corresponding HDM errors match the linear-PROM level on the checked points (e.g., 0.410% and 0.450%). This supports that the PG assembly/solve is consistent; the practical PG gap in Table 1 and Table 2 is driven by closure-map accuracy, not by the PG implementation itself.

5.5 Interpretation of the Case 2 $n = 20$ behavior

At first glance, Case 2 with $n = 20$ may seem as if it should be easier than $n = 10$, because the ANN output size drops from $\bar{n} = 141$ to $\bar{n} = 131$. However, in our setup, the main difficulty is not only output dimension; it is generalization of the map $(\mu, t) \mapsto \bar{\mathbf{q}}$ across the domain with only 9 baseline trajectories.

For Case 2 specifically, $\bar{\mathbf{q}}$ does not depend on the online state $\mathbf{q}$. This means the model cannot correct truncated content based on instantaneous reduced-state deviations during the solve. Changing from $n = 10$ to $n = 20$ changes the retained/truncated split and therefore changes which dynamics must be represented by the standalone map $\mathcal{M}(\mu, t)$. With limited baseline data, this can degrade off-grid robustness even if the output dimension is smaller.

In the direct baseline $n = 20$ training, this appears as severe off-grid degradation. A transfer check (reuse baseline $n = 10$ model, then trim outputs for $n = 20$) substantially improves off-grid errors, indicating that the baseline issue is largely tied to training quality/data coverage of the native $n = 20$ map rather than a fundamental inconsistency of the $n = 20$ online solve itself.

5.6 Coefficient-space error analysis and source decomposition

Unless stated otherwise, the coefficient-space diagnostics below use the linear $n_{\text{tot}} = 151$ PROM coefficients as reference. Baseline plots include the extended Case 2 family (PG, PRNN, and $n = 20$ variants), while enriched plots include all available enriched runs.

To isolate reduced-coordinate effects, we use the linear n_{tot} -PROM coefficients as reference:

$$\mathbf{q}^{\text{ref}}(t; \mu) := \mathbf{q}_N^{\text{Linear PROM}}(t; \mu).$$

For each model m , with coefficients $\mathbf{q}^{(m)}$, we define per-mode trajectory errors

$$E_i^{\text{abs}}(m; \mu) = \left\| q_i^{\text{ref}} - q_i^{(m)} \right\|_2, \quad E_i^{\text{rel}}(m; \mu) = \frac{E_i^{\text{abs}}(m; \mu)}{\left\| q_i^{\text{ref}} \right\|_2}.$$

Here, the norm is taken over time and i indexes POD coefficients.

We split reduced coordinates as $\mathbf{q}_N = [\mathbf{q}; \bar{\mathbf{q}}]$, where $\mathbf{q}$ is retained and $\bar{\mathbf{q}}$ is truncated (implementation aliases: $\mathbf{q}_p, \mathbf{q}_s$). We define **Source 1** (retained-coordinate mismatch, i.e. mismatch

in the solved LSPG unknown $\mathbf{q}$) and total truncated mismatch as

$$\mathbf{e}_q^{(m)} := \mathbf{q}^{\text{ref}} - \mathbf{q}^{(m)}, \quad \eta_q^{(m)} := \frac{\|\mathbf{e}_q^{(m)}\|_F}{\|\mathbf{q}^{\text{ref}}\|_F},$$

$$\mathbf{e}_{\bar{q},\text{tot}}^{(m)} := \bar{\mathbf{q}}^{\text{ref}} - \bar{\mathbf{q}}^{(m)}, \quad \eta_{\bar{q}}^{(m)} := \frac{\|\mathbf{e}_{\bar{q},\text{tot}}^{(m)}\|_F}{\|\bar{\mathbf{q}}^{\text{ref}}\|_F}.$$

Table 5 summarizes means over the three reported points.

Table 5: Mean relative coefficient errors (%) over $\boldsymbol{\mu}^{(v)}, \boldsymbol{\mu}^{(1)}, \boldsymbol{\mu}^{(2)}$, using linear n_{tot} -PROM coefficients as reference.

Method	Baseline η_q [%]	Baseline $\eta_{\bar{q}}$ [%]	Enriched η_q [%]	Enriched $\eta_{\bar{q}}$ [%]
Case 1 ANN	0.80	7.55	0.67	6.08
Case 2 ANN	1.54	9.88	1.62	4.49
Case 2 PRNN	1.64	10.17	—	—
Case 3 ANN	0.71	6.40	0.29	2.69
Data-driven	2.04	11.69	1.08	7.46

Error-source decomposition for closure models. For Cases 1–3, the truncated-coordinate error $\mathbf{e}_{\bar{q},\text{tot}}$ can be decomposed as follows. The Case 2 PRNN model uses the same Case 2 functional structure $\bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t)$, so its decomposition has the same form (no Source 3 term); we report it as an additional baseline reference in Table 6. There is no independent “Source 1” term for $\bar{\mathbf{q}}$ because $\bar{\mathbf{q}}$ is not solved as an independent unknown in these closure models. Source 1 acts on $\bar{\mathbf{q}}$ only indirectly through Source 3 when the closure depends on $\mathbf{q}$ (Cases 1 and 3). For Case 2, Source 3 is identically absent.

Case 1:

$$\mathbf{e}_{\bar{q},\text{tot}}^{(1)} = \bar{\mathbf{q}}^{\text{ref}} - \bar{\mathbf{q}}^{(1)} = \underbrace{\left(\bar{\mathbf{q}}^{\text{ref}} - \mathcal{N}(\mathbf{q}^{\text{ref}})\right)}_{\text{Source 2: map mismatch}} + \underbrace{\left(\mathcal{N}(\mathbf{q}^{\text{ref}}) - \mathcal{N}(\mathbf{q}^{(1)})\right)}_{\text{Source 3: continuity / sensitivity}}.$$

Case 2:

$$\mathbf{e}_{\bar{q},\text{tot}}^{(2)} = \bar{\mathbf{q}}^{\text{ref}} - \bar{\mathbf{q}}^{(2)} = \underbrace{\left(\bar{\mathbf{q}}^{\text{ref}} - \mathcal{M}(\boldsymbol{\mu}, t)\right)}_{\text{Source 2: map mismatch}}.$$

Case 3:

$$\mathbf{e}_{\bar{q},\text{tot}}^{(3)} = \bar{\mathbf{q}}^{\text{ref}} - \bar{\mathbf{q}}^{(3)} = \underbrace{\left(\bar{\mathbf{q}}^{\text{ref}} - \mathcal{H}(\mathbf{q}^{\text{ref}}, \boldsymbol{\mu}, t)\right)}_{\text{Source 2: map mismatch}} + \underbrace{\left(\mathcal{H}(\mathbf{q}^{\text{ref}}, \boldsymbol{\mu}, t) - \mathcal{H}(\mathbf{q}^{(3)}, \boldsymbol{\mu}, t)\right)}_{\text{Source 3: continuity / sensitivity}}.$$

For Cases 1 and 3, Source 3 can be interpreted as propagation of Source 1 through a $\mathbf{q}$ -dependent closure. Under an additional regularity assumption, this propagation can be bounded: if the closure map is locally Lipschitz in the neighborhood visited by the trajectories, with local constant $L_{\mathcal{F}}$, then

$$\|\text{Source 3}\|_F \leq L_{\mathcal{F}} \|\mathbf{e}_q\|_F.$$

This inequality is used here as a qualitative interpretation aid only: we do not estimate $L_{\mathcal{F}}$ from data, nor claim tightness of the bound. Source 1 is explicitly reported by η_q in Table 5, while Table 6 resolves Source 2 and Source 3 on truncated coefficients.

Table 6 reports mean relative magnitudes of these terms over the three points.

Table 6: Mean relative decomposition terms on truncated coefficients (%) over $\boldsymbol{\mu}^{(v)}, \boldsymbol{\mu}^{(1)}, \boldsymbol{\mu}^{(2)}$.

Stage	Case	Source 2 [%]	Source 3 [%]	Total $\eta_{\bar{q}}$ [%]
Baseline	Case 1	5.50	6.13	7.55
Baseline	Case 2	9.88	—	9.88
Baseline	Case 2 PRNN	10.17	—	10.17
Baseline	Case 3	3.58	5.84	6.40
Enriched	Case 1	3.44	5.33	6.08
Enriched	Case 2	4.49	—	4.49
Enriched	Case 3	2.40	1.68	2.69

Figures 5–10 show per-index absolute and relative coefficient errors for all three reported parameter points. The display order is parameter-first; for each parameter point, we report baseline first and enriched second. Baseline curves include Case 1, Case 2, Case 2 PRNN, Case 2 PG Oracle, Case 2 $n = 20$ (native), Case 2 $n = 20$ (trimmed transfer), Case 3, and data-driven; enriched curves include the available enriched models (no enriched PRNN or trimmed-transfer run in this campaign).

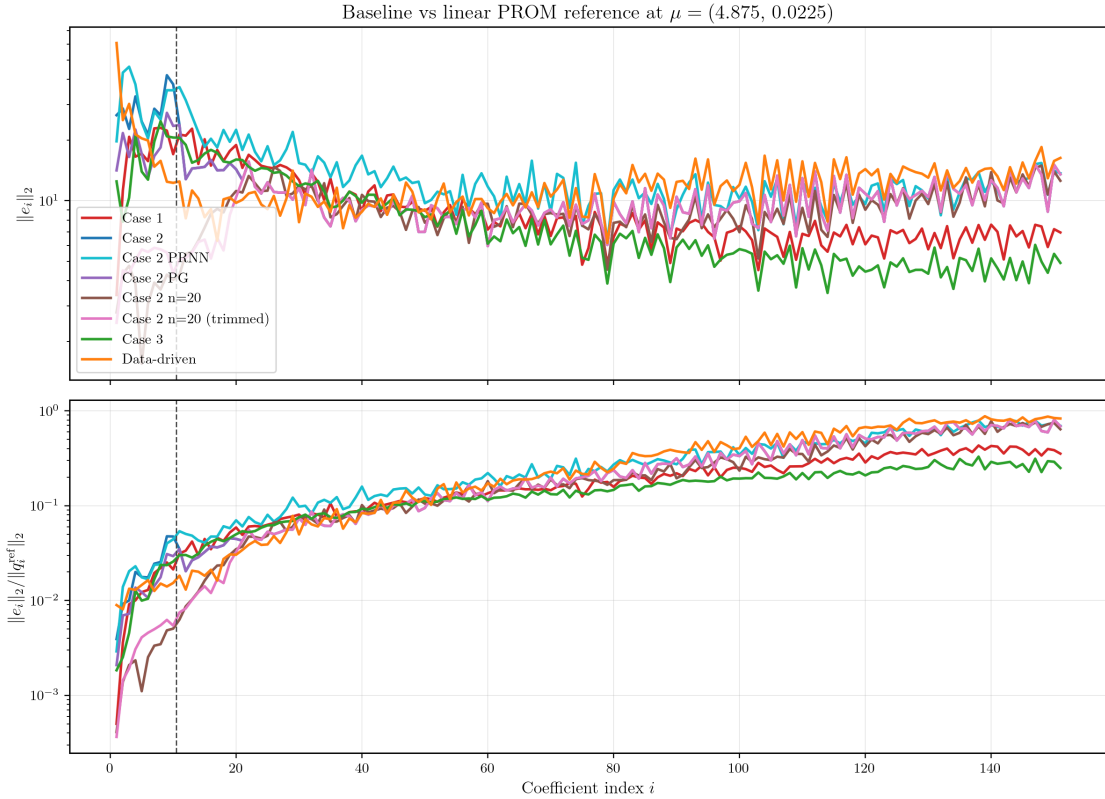


Figure 5: Baseline stage: per-index absolute and relative coefficient errors vs linear n_{tot} -PROM reference at $\boldsymbol{\mu}^{(v)} = (4.875, 0.0225)$, including Case 2 PRNN, Case 2 PG Oracle, Case 2 $n = 20$ (native), and Case 2 $n = 20$ (trimmed transfer).

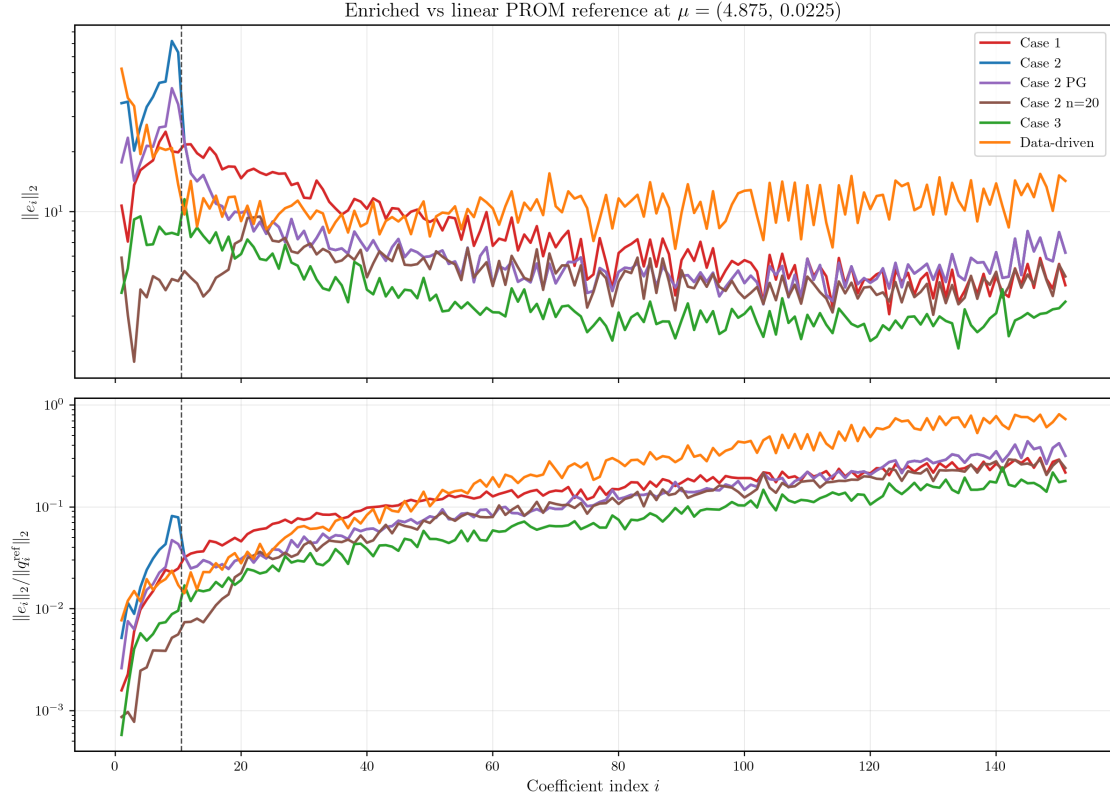


Figure 6: Enriched stage: per-index absolute and relative coefficient errors vs linear n_{tot} -PROM reference at $\mu^{(v)} = (4.875, 0.0225)$, including Case 2 PG Oracle and Case 2 $n = 20$ (native).

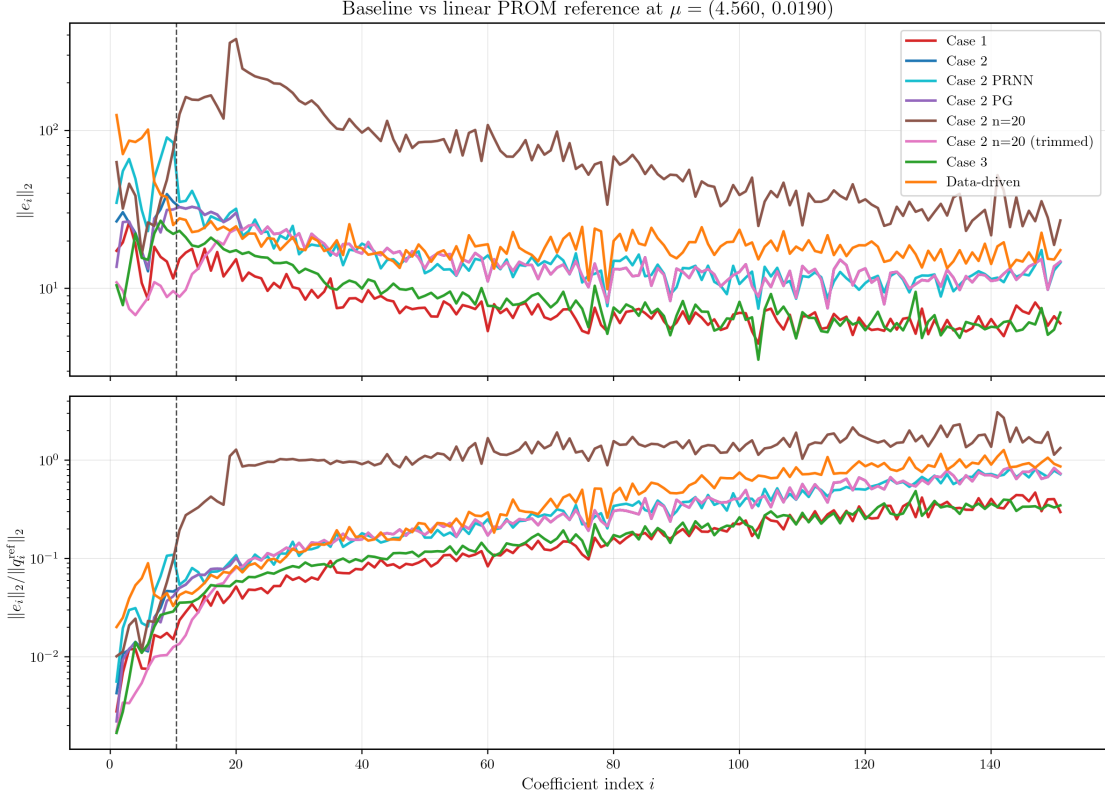


Figure 7: Baseline stage: per-index absolute and relative coefficient errors at $\mu^{(1)} = (4.56, 0.019)$, including Case 2 PRNN, Case 2 PG Oracle, Case 2 $n = 20$ (native), and Case 2 $n = 20$ (trimmed transfer).

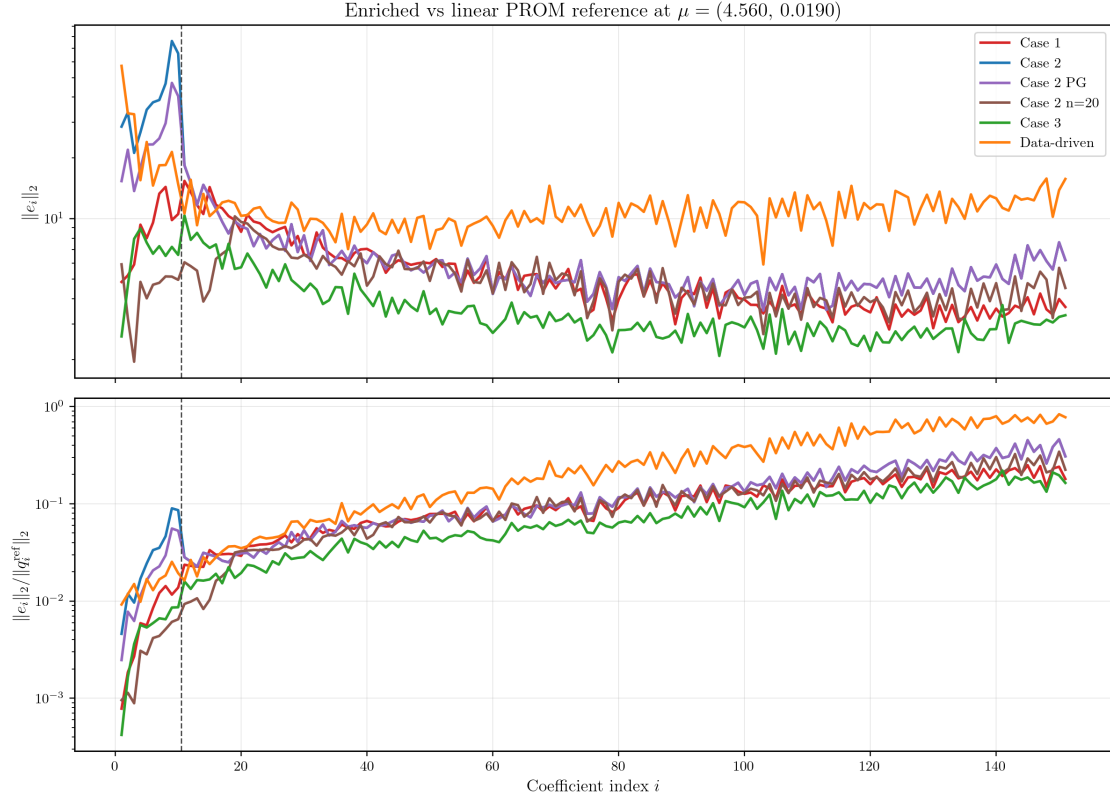


Figure 8: Enriched stage: per-index absolute and relative coefficient errors at $\boldsymbol{\mu}^{(1)} = (4.56, 0.019)$, including Case 2 PG Oracle and Case 2 $n = 20$ (native).

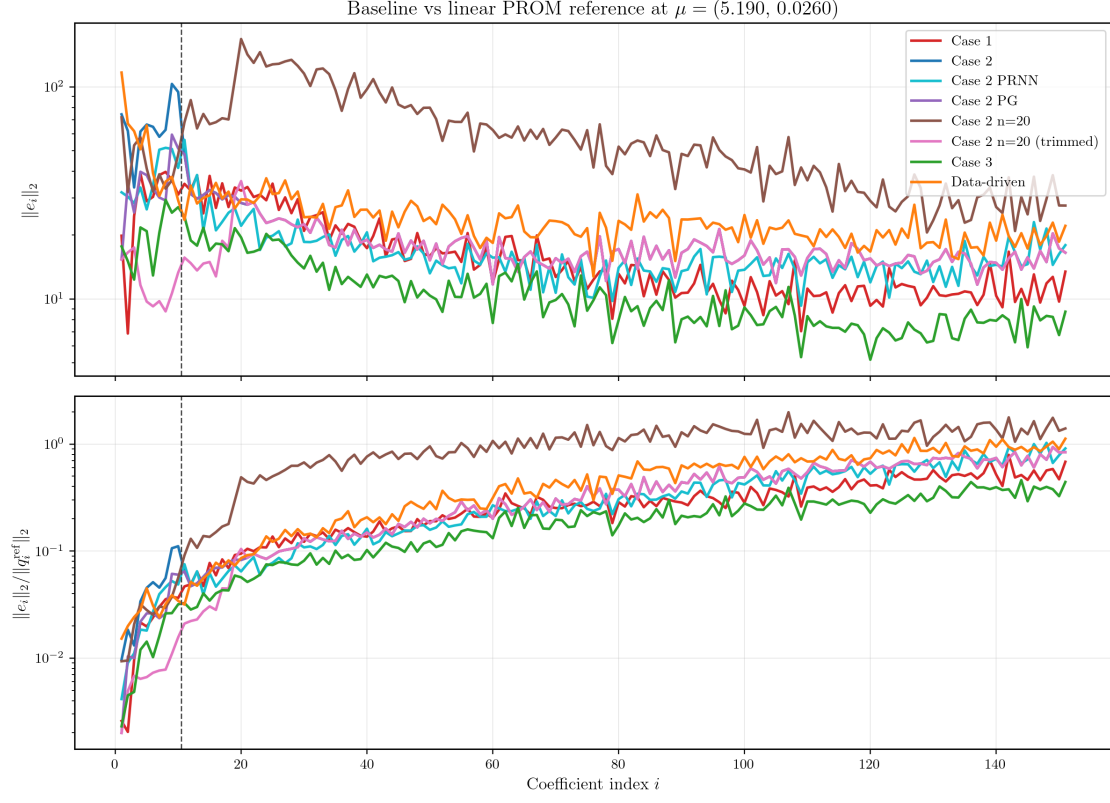


Figure 9: Baseline stage: per-index absolute and relative coefficient errors at $\boldsymbol{\mu}^{(2)} = (5.19, 0.026)$, including Case 2 PRNN, Case 2 PG Oracle, Case 2 $n = 20$ (native), and Case 2 $n = 20$ (trimmed transfer).

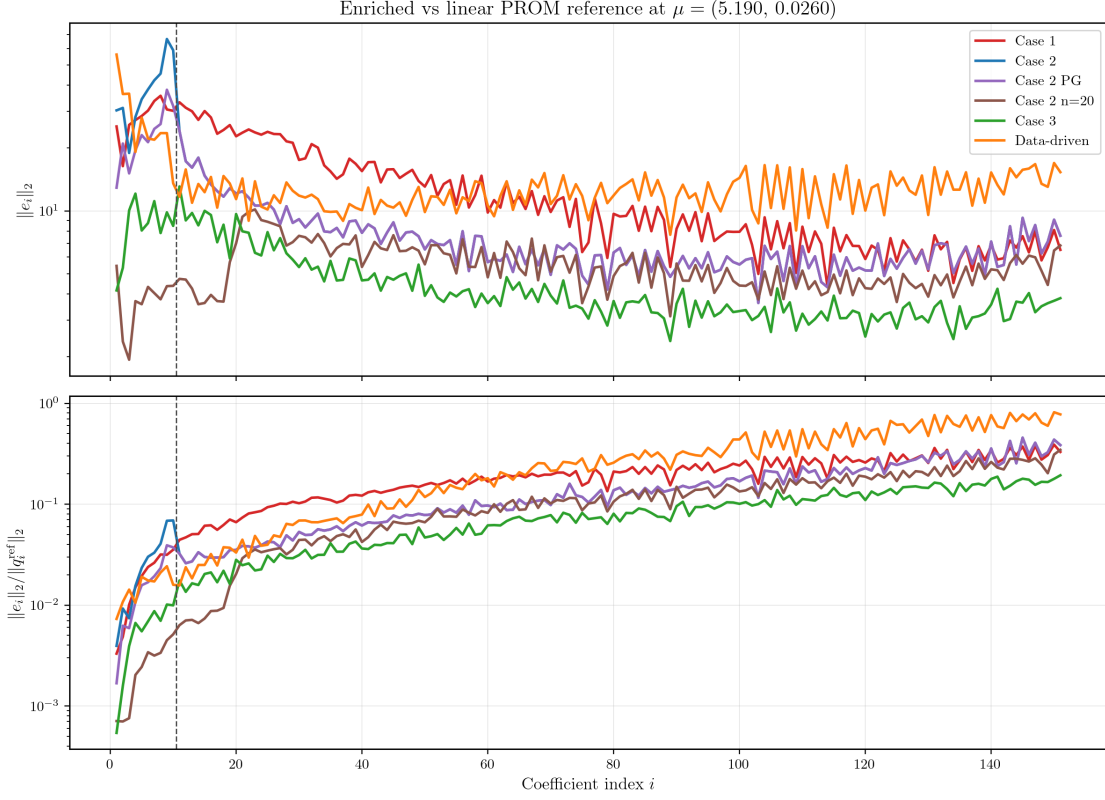


Figure 10: Enriched stage: per-index absolute and relative coefficient errors at $\boldsymbol{\mu}^{(2)} = (5.19, 0.026)$, including Case 2 PG Oracle and Case 2 $n = 20$ (native).

5.7 Qualitative comparison: HDM vs all models

Figures 11 and 12 compare HDM (black) against ANN-ROM variants, ordered as $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, and $\boldsymbol{\mu}^{(2)}$. The baseline panel includes Case 2 PRNN in addition to Case 1/2/3 and data-driven; the enriched panel reports the available enriched models.

Baseline (9-point training, PROM online): HDM vs Case1/Case2/Case2-PRNN/Case3/Data-driven

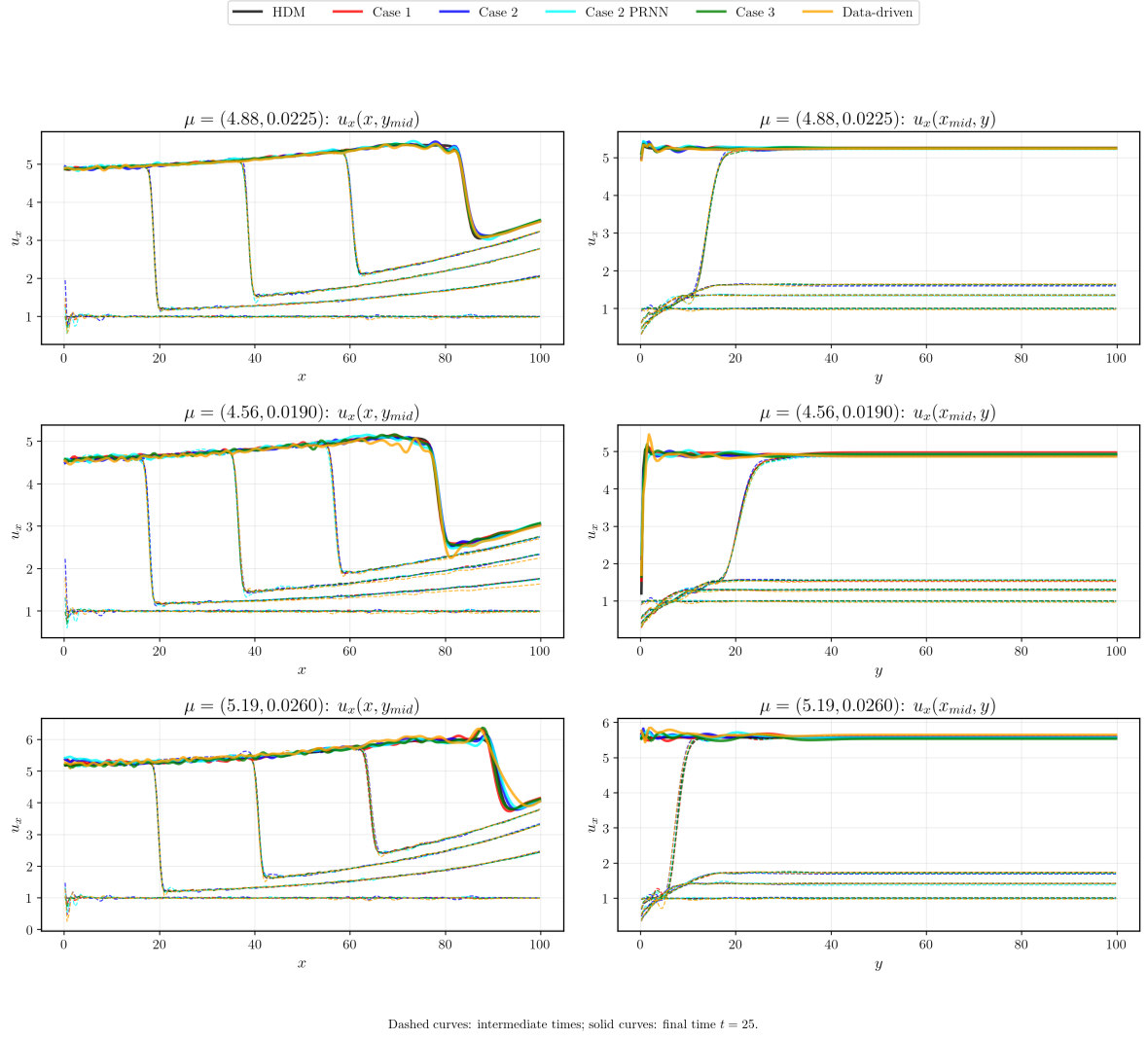


Figure 11: Baseline models, PROM online solve: HDM vs Case 1, Case 2, Case 2 PRNN, Case 3, and data-driven at $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$.

Enriched (9+20 training, PROM online): HDM vs Case1/Case2/Case3/Data-driven

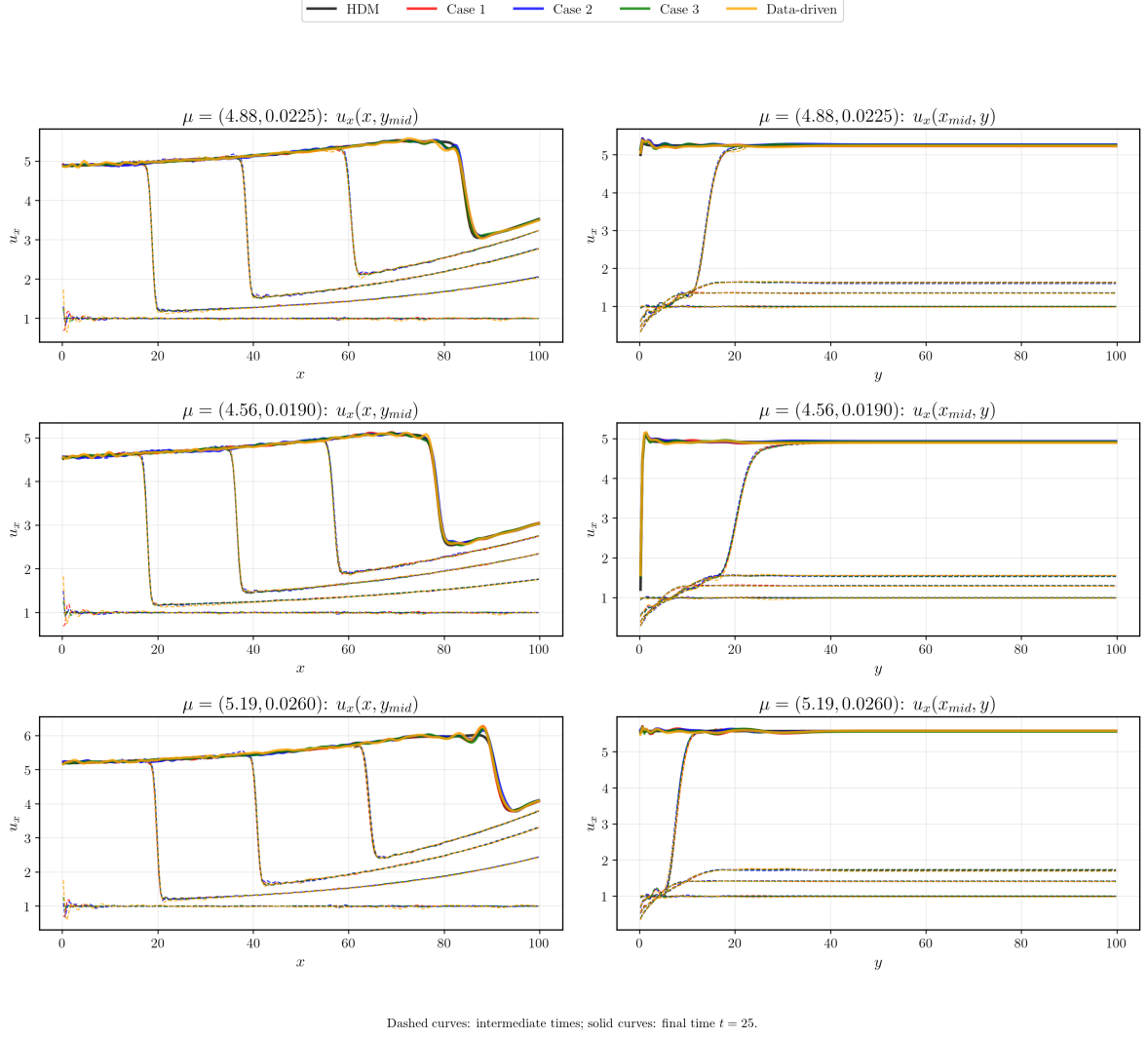


Figure 12: Enriched models, PROM online solve: HDM vs available enriched models (Case 1/2/3 and data-driven) at $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$.

5.8 Time-Resolved Coefficient Error Heatmaps

To visualize error propagation in time, we report coefficient-index vs time heatmaps at $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$, with linear n_{tot} -PROM coefficients as reference. Baseline heatmaps include all available baseline variants (Case 1, Case 2, Case 2 PRNN, Case 2 PG Oracle, Case 2 $n = 20$ native, Case 2 $n = 20$ trimmed transfer, Case 3, data-driven). Enriched heatmaps include all available enriched variants (Case 1, Case 2, Case 2 PG Oracle, Case 2 $n = 20$ native, Case 3, data-driven).

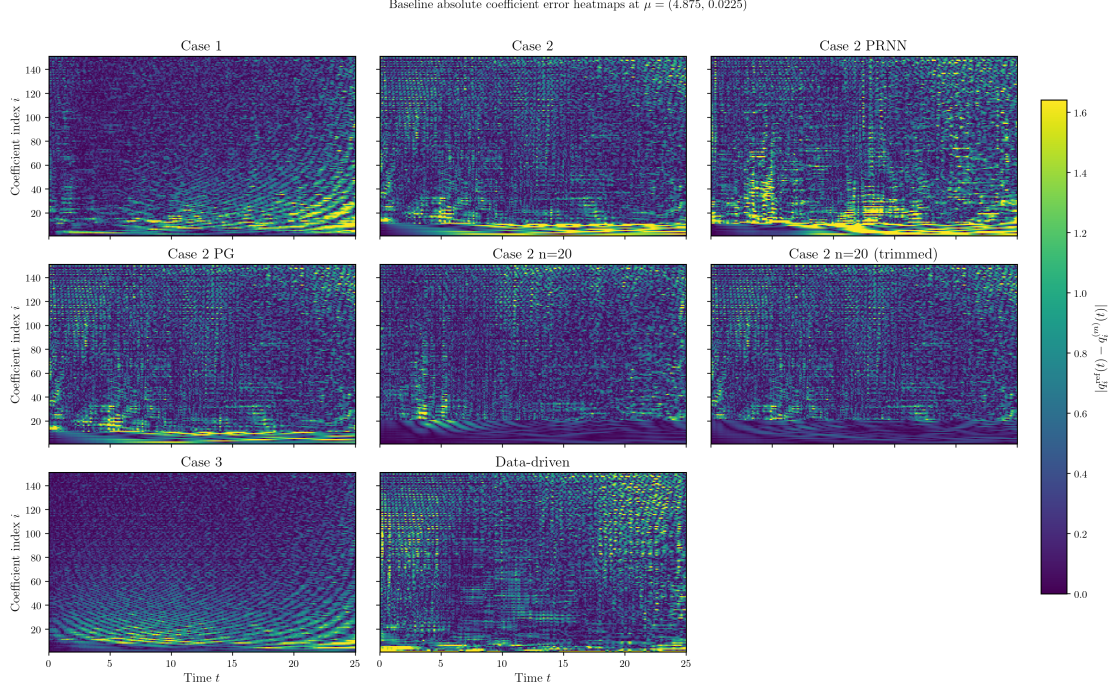


Figure 13: Baseline stage: absolute coefficient error heatmaps $\left|q_i^{\text{ref}}(t) - q_i^{(m)}(t)\right|$ for all available baseline models.

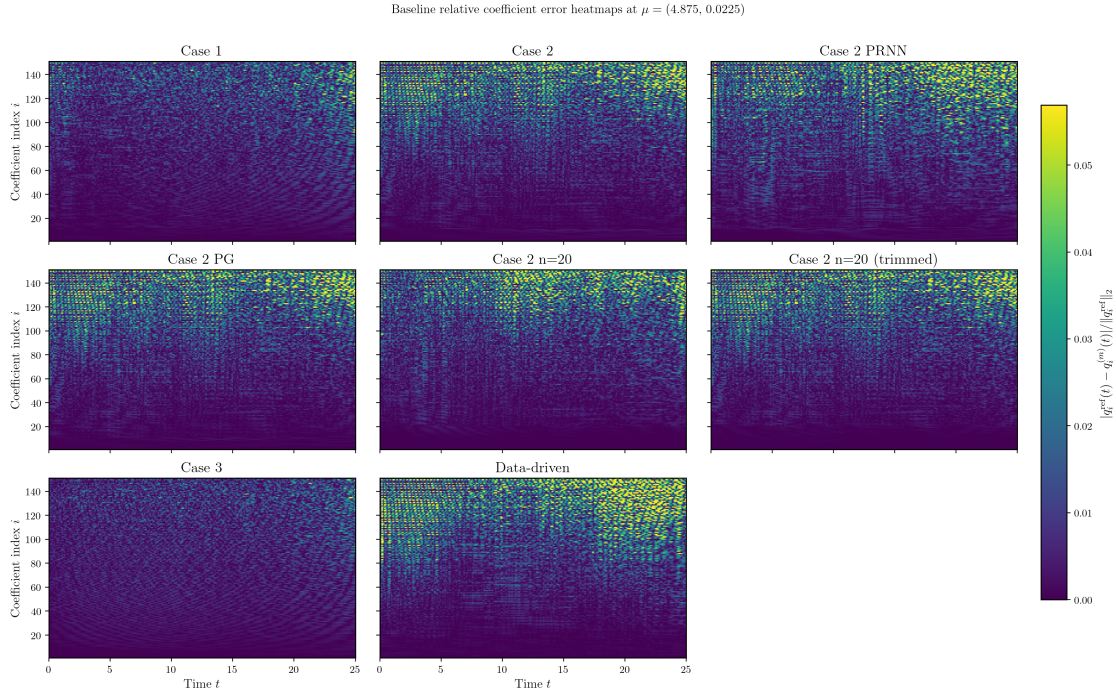


Figure 14: Baseline stage: relative coefficient error heatmaps for all available baseline models, using normalization by $\|q_i^{\text{ref}}\|_2$.

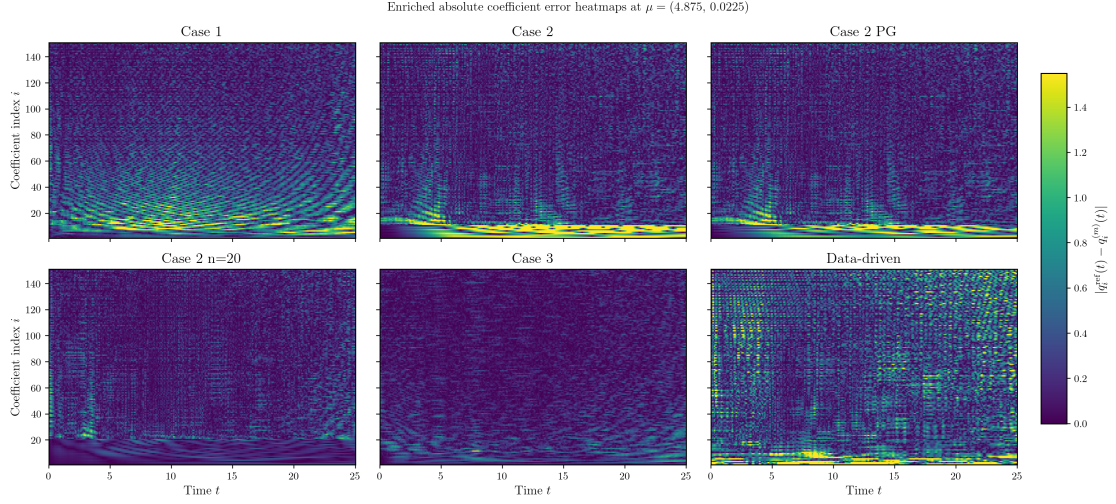


Figure 15: Enriched stage: absolute coefficient error heatmaps at $\mu^{(v)}$ for all available enriched models.

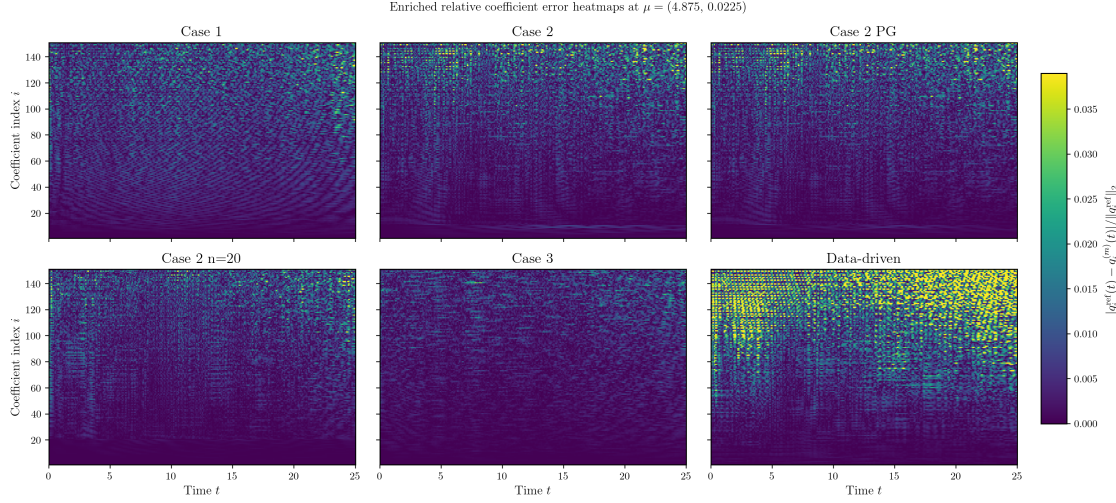


Figure 16: Enriched stage: relative coefficient error heatmaps at $\mu^{(v)}$ for all available enriched models.

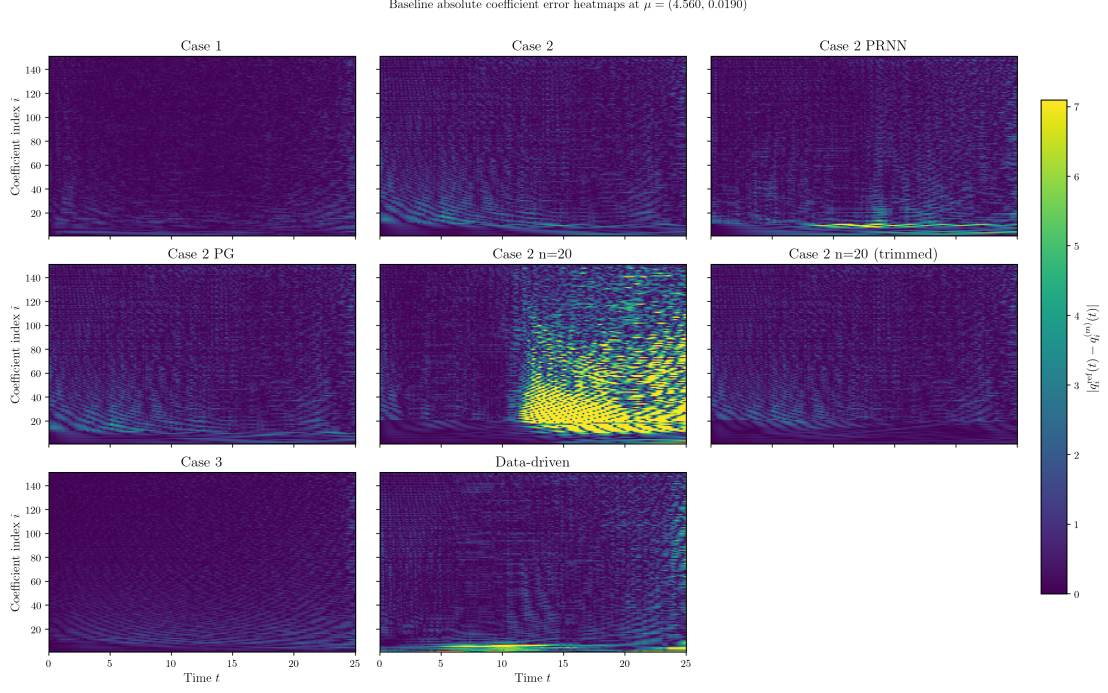


Figure 17: Baseline stage: absolute coefficient error heatmaps at $\mu^{(1)} = (4.56, 0.019)$ for all available baseline models.

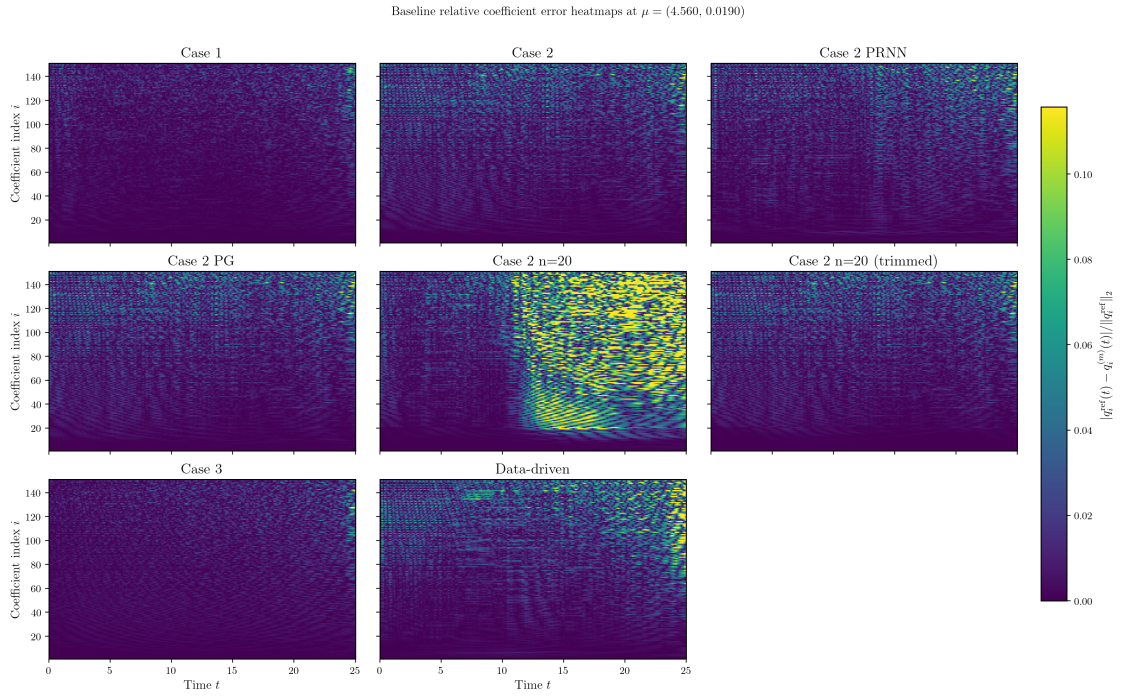


Figure 18: Baseline stage: relative coefficient error heatmaps at $\mu^{(1)} = (4.56, 0.019)$ for all available baseline models.

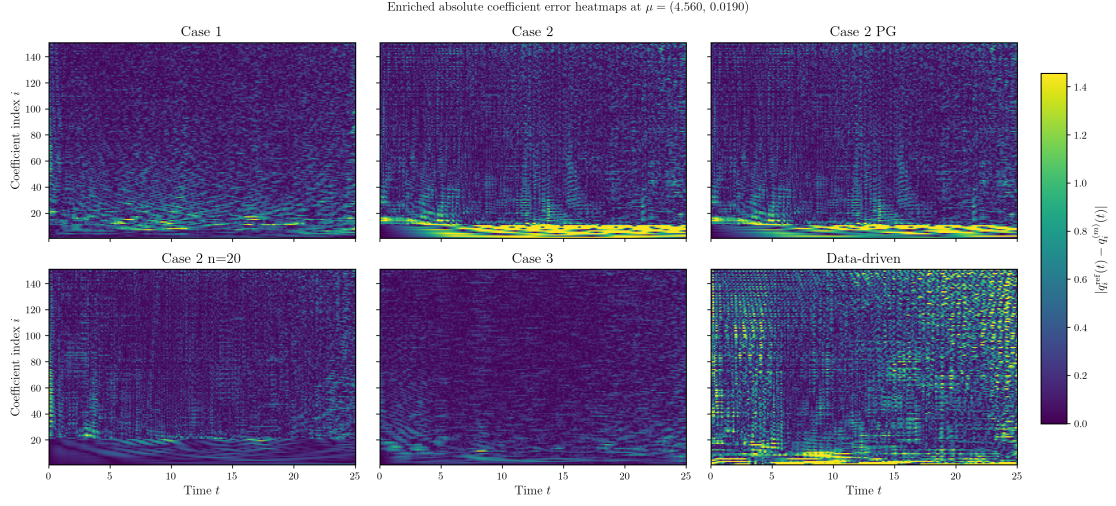


Figure 19: Enriched stage: absolute coefficient error heatmaps at $\mu^{(1)} = (4.56, 0.019)$ for all available enriched models.

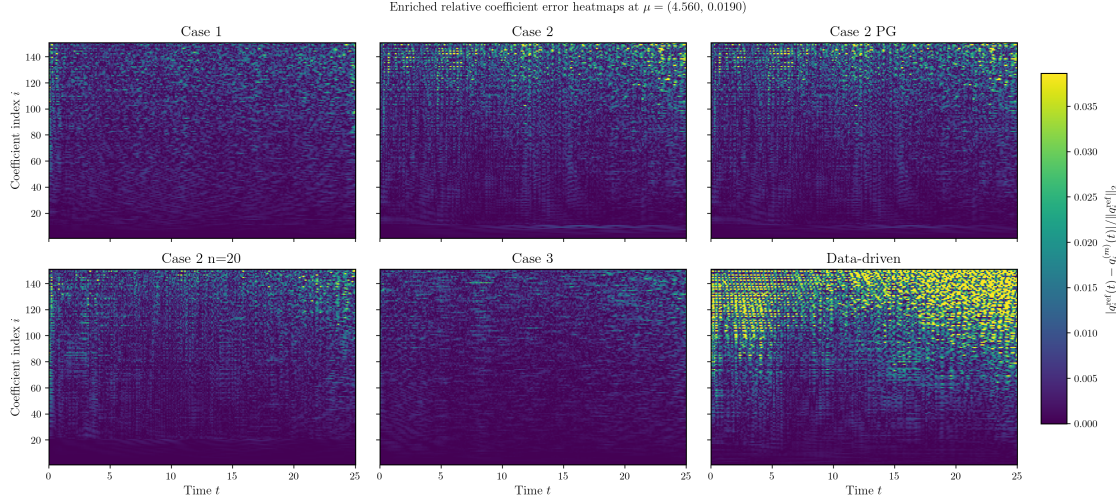


Figure 20: Enriched stage: relative coefficient error heatmaps at $\mu^{(1)} = (4.56, 0.019)$ for all available enriched models.

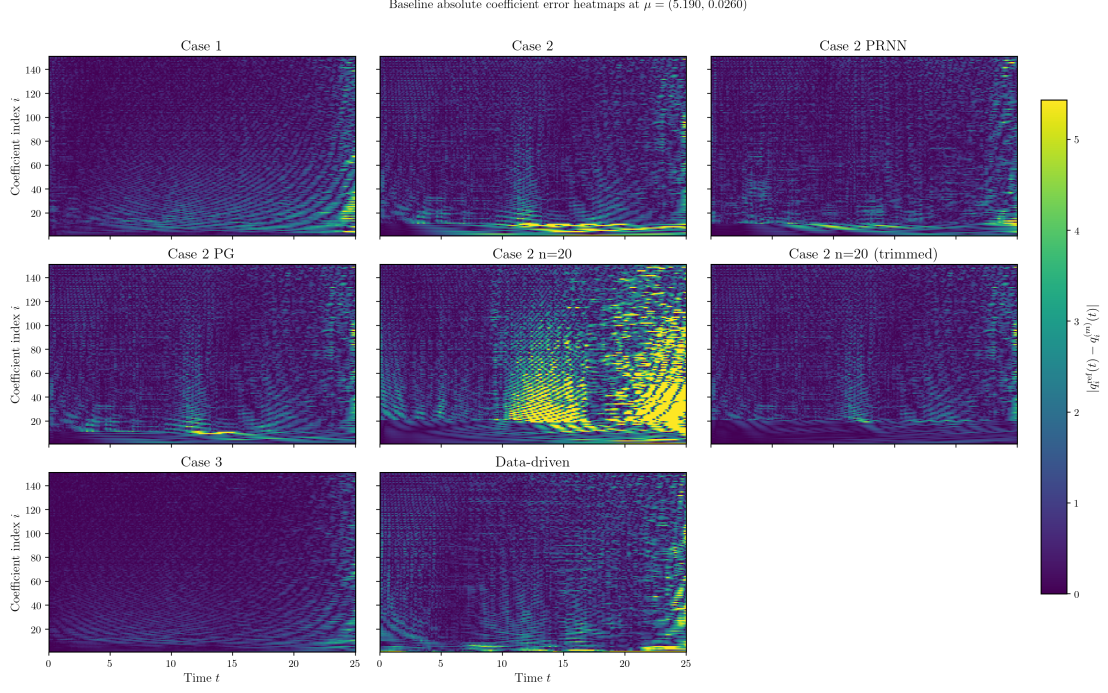


Figure 21: Baseline stage: absolute coefficient error heatmaps at $\mu^{(2)} = (5.19, 0.026)$ for all available baseline models.

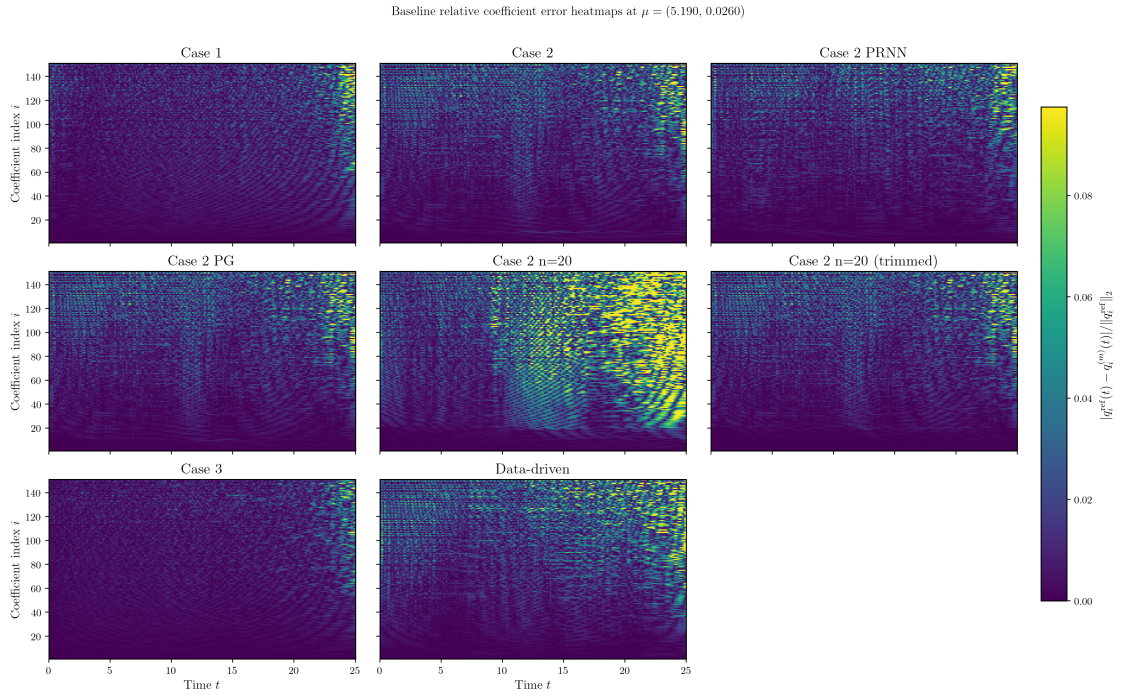


Figure 22: Baseline stage: relative coefficient error heatmaps at $\mu^{(2)} = (5.19, 0.026)$ for all available baseline models.

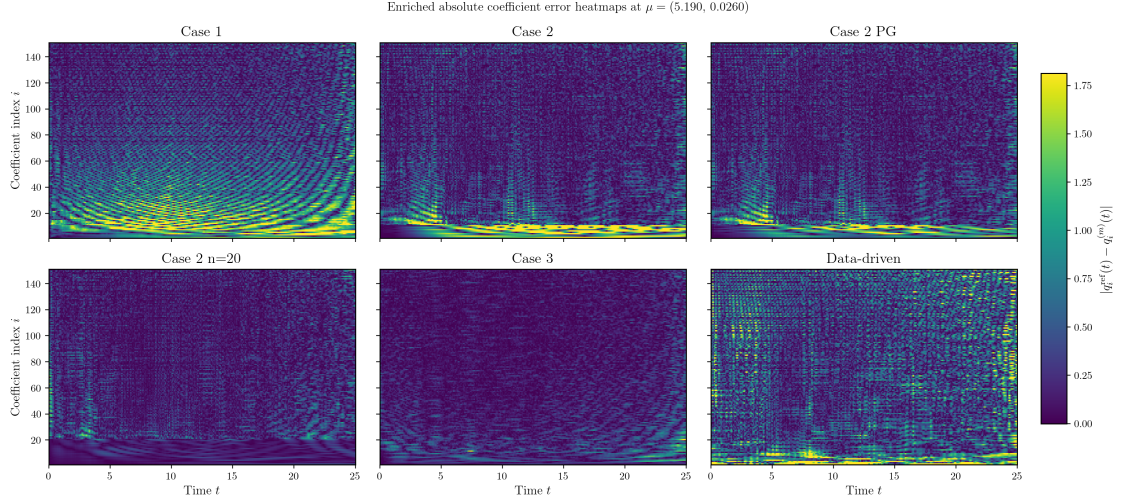


Figure 23: Enriched stage: absolute coefficient error heatmaps at $\mu^{(2)} = (5.19, 0.026)$ for all available enriched models.

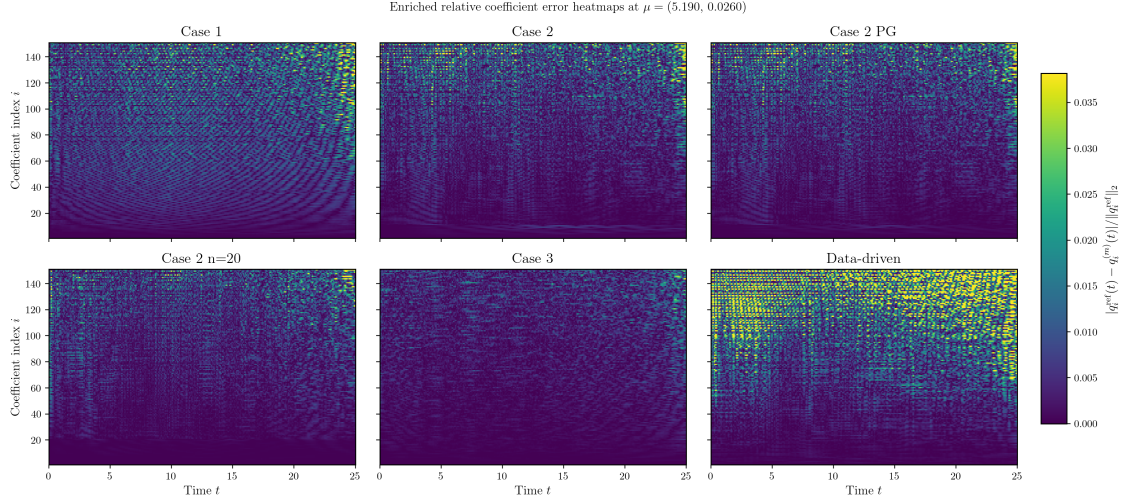


Figure 24: Enriched stage: relative coefficient error heatmaps at $\mu^{(2)} = (5.19, 0.026)$ for all available enriched models.